# IF A DIRECT SUM OF TWO OBJECTS OF THE DERIVED CATEGORY OF A RINGED SITE HAS TOR AMPLITUDE IN A GIVEN INTERVAL, SO DOES EACH SUMMAND

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: Same-name restatement check: the label lemma-summands-tor-amplitude also occurs in the chapters cohomology.tex (ringed spaces) and more-algebra.tex (rings); both are special cases of the ringed-site statement, which is the more general one counted here.

*Editorial note.* The label lemma-summands-tor-amplitude defines the statement; the counted proof is the proof environment immediately following the lemma in the source.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

*Allowed standard prerequisites (editorial).*

- Tor amplitude in an interval. Same chapter, the definition of tor amplitude of an object of D(O) via the vanishing of Tor_O^q(K, F) outside an interval for all O-modules F. Supplies the additivity of the Tor functors that makes the one-line proof valid.

**Lemma 0.1.** *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $a, b \in \mathbf{Z}$. For $K$, $L$ objects of $D(\mathcal{O})$ if $K \oplus L$ has tor amplitude in $[a, b]$ so do $K$ and $L$.*

*Proof.* Clear from the fact that the Tor functors are additive. □

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